

d -Translated Unit Sensitive Primes

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Abstract

A nonnegative integer n is d -translated unit sensitive when appending d zeros to the right of the number and changing the unit digit to any possible unit digit (so long as the resulting number is not n) results in a composite number. We find an arithmetic progression of nonnegative integers that are d -translated unit sensitive for any nonnegative integer d . This arithmetic progression is constructed such that there exists k consecutive primes in the progression for any positive integer k . We also give the first known explicit example of a prime with this property. We also find an arithmetic progression of nonnegative integers that are both d -translated unit sensitive for any nonnegative integer d and Brier numbers. This arithmetic progression is also constructed such that there exists k consecutive primes in the progression for any positive integer k .

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1. Introduction

In 1950, Paul Erdős introduced the technique of covering systems as a constructive method to answer questions dealing with infinitely many integers (see [6]). Since then, covering systems have been used frequently as a proof technique (see [8], [11], and [12]) while also having its own questions to be answered (see [1], [5], and [16]). This paper will use covering systems in a way similar to that of Erdős in [6]. Thus, we first define a covering system.

Definition 1 (Covering System). *A covering system (or covering) is a finite set of congruences,*

$$\{x \equiv a_1 \pmod{b_1}, \quad x \equiv a_2 \pmod{b_2}, \quad \dots, \quad x \equiv a_n \pmod{b_n}\},$$

where n is a positive integer, each a_i is an integer, and each b_i is a positive integer such that every integer satisfies at least one congruence in the set.

To see how a covering can be useful, consider the first use by Erdős in 1950 (see [6]). Erdős created coverings to help answer a question of Romonoff (which was about a conjecture of De Polignac in [19]), namely, are there infinitely many odd integers that cannot be expressed in the form $2^k + p$ for some p is a prime and k some nonnegative integer.

To prove there are infinitely many (and disprove this conjecture), Erdős sought to construct an arithmetic progression of composite odd integers of the form $2^k - x$. He began by constructing a covering system for all choices of $k \in \mathbb{Z}$ (the first column of Table 1).

Table 1: Erdős Covering System

Congruence Classes for k	Congruence Classes for x
$0 \pmod{2}$	$1 \pmod{3}$
$0 \pmod{3}$	$1 \pmod{7}$
$1 \pmod{4}$	$2 \pmod{5}$
$3 \pmod{8}$	$8 \pmod{17}$
$7 \pmod{12}$	$11 \pmod{13}$
$23 \pmod{24}$	$121 \pmod{241}$

From these congruence classes making up the covering system, he carefully selected congruence classes for x to satisfy so that $2^k - x$ is divisible by some prime (the second column of the Table 1). For example, consider the second row of Table 1. When $k \equiv 0 \pmod{3}$ and $x \equiv 1 \pmod{7}$, we have, for the non-negative integer n ,

$$x - 2^k = x - 2^{0+3n} = x - 2^0 \cdot 2^{3n} \equiv 1 - 1 \cdot 1 \equiv 0 \pmod{7}.$$

Thus, for $k \equiv 0 \pmod{3}$ and $x \equiv 1 \pmod{7}$, it will always be true that 7 divides $x - 2^k$. Since the moduli in the congruence classes for x in the table are all unique primes, so long as $x - 2^k$ is not equal to any of the prime moduli, $x - 2^k$ will be composite for $x \equiv 7629217 \pmod{11184810}$ by the Chinese Remainder Theorem. Written as an arithmetic progression, for any nonnegative integer m , this is

$$x = 11184810m + 7629217.$$

While many of the results using coverings as a constructive proof technique have followed this same general structure, several conjectures and results pertaining to the construction of covering systems have been a topic of recent works as well. One

famous open question that was proposed by Selfridge is “Can a covering system be constructed with unique odd moduli?” Another famous problem, proposed by Erdős in [6], known as the minimum modulus problem was tackled by Bob Hough in 2015 (see [16]). Hough showed the minimum modulus of a covering system must be less than 10^{16} . In the eleven years since this finding, the distortion method has been used to improve this bound to 616000. However, the explicit minimum modulus of a covering system is still an open problem.

Some well-known results relating to covering systems and the constructions used in this paper are that of Sierpinski, Riesel, and Clavier. Sierpinski used covering systems to prove that there were infinitely many positive odd integers k that $k \cdot 2^n + 1$ is composite for all non-negative integers n and that for any non-negative integer m ,

$$k = 36893488147419103230m + 15511380746462593381$$

has this property (see [22]). Consequently, integers of this form are referred to as *Sierpinski numbers*. Likewise, Riesel proved that there were infinitely many positive odd integers k that $k \cdot 2^n - 1$ is composite for all non-negative integers n and that for any non-negative integer m ,

$$k = 11184810m + 509203$$

has this property (see [20]). Consequently, integers of this form are referred to as *Riesel numbers*. Similarly, Clavier found the smallest known example of a *Brier number* (a number that is both a Sierpinski number and Riesel number) by constructing an arithmetic progression where for any nonnegative integer m

$$k = 3316923598096294713661 + 3770214739596601257962594704110m$$

has these two properties (see [15] and [8]). Filaseta and Juillerat showed that this construction contains infinitely many Brier primes. For other recent results pertaining to Sierpinski, Riesel, and Brier numbers see: [2], [3], [4], [7], [11], [14], [23], and [24]. The results of this paper are a consequence of similar techniques used in the aforementioned paper. We first define some key terms.

Definition 2 (Unit Sensitive). *Let any positive integer be unit sensitive if changing the unit digit to any other base 10 digit results in a composite number.*

The main result of this paper pertains to prime numbers, so we will be focusing specifically on unit sensitive primes. Consider the prime 97. Changing the unit digit 7 results in one of the following integers:

$$90, 91, 92, 93, 94, 95, 96, 98, 99.$$

All of these integers are composite. Thus, 97 is a unit sensitive prime. This is not a particularly rare occurrence. In fact, it is possible to check if a prime is unit

sensitive by using only 9 primality checks. Thus, consider the following natural extension.

Definition 3 (*d*-Translated Unit Sensitive). *For any nonnegative integer d , a positive integer is d -translated unit sensitive if appending d zeros to the right of the number and changing the unit digit to any possible unit digit results in a composite number.*

Note that in the case $d = 0$, the number is unit sensitive.

To illustrate this construction consider the same prime from before 97. To see if 97 is 1-translated unit sensitive, we need only consider the following integers:

$$970, 971, 972, 973, 974, 975, 976, 977, 978, 979.$$

We say 97 is not 1-translated unit sensitive since 971 and 977 are prime. By similar argument, it is easy to show the first 1-translated unit sensitive prime is 53. By repeating this procedure, it is easy to show the first 1, 2-translated unit sensitive prime is 89. For simplicity, we will say 89 is the first $[1, 2]$ -translated unit sensitive prime. In fact, as seen in Table 2 below, none of the first 10^6 primes are $[1, 4381]$ -translated unit sensitive and in Table 3 below none of the first 10^6 primes are $[0, 3191]$ -translated unit sensitive.

Table 2: Computational Results $d \geq 1$

# of primes, p	Max d	p using Max d
10^2	9	443
10^3	38	7829
10^4	269	52121
10^5	513	390893
10^6	4380	2049797

Table 3: Computational Results $d \geq 0$

# of primes, p	Max d	p using Max d
10^2	5	113
10^3	11	5693
10^4	51	37379
10^5	163	951161
10^6	3190	10721819

Similar to testing for unit sensitivity, d -translated unit sensitivity requires only 10 primality checks for each choice of d . Thus, a computer can be used to check these properties in a reasonable amount of time. We further restrict this property to

$[0, \infty)$ -translated unit sensitive primes. Since there are infinitely choices of d in this case, a simple computer algorithm is not feasible for finding such primes. In fact, Table 3 above suggests that there may not be any $[0, \infty)$ -translated unit sensitive primes. Contrary to this suspicion, we prove the following results and corollaries.

Theorem 1. *There are infinitely many primes that are d -translated unit sensitive for any nonnegative integer d (or $[0, \infty)$ -translated unit sensitive). In fact, the arithmetic progression, $Am + B$ for any nonnegative integer m , contains infinitely many such primes where*

$$A = 124703827160493827160493827160493827160493827160493827035790 \text{ and}$$

$$B = 41459060189171787548442999328384678040412832671445258454633.$$

Since B is prime, it is the first known example of a $[0, \infty)$ -translated unit sensitive prime. The next result adds the condition of being a Brier number to the arithmetic progression.

Theorem 2. *There are infinitely many primes that are d -translated unit sensitive for any nonnegative integer d (or $[0, \infty)$ -translated unit sensitive) and are Brier numbers. In fact, the arithmetic progression, $Am + B$ for any nonnegative integer m , contains infinitely many such primes where*

$$A = 105882605432768363742137390170084901366252883723703759364986698321995$$

$$2824339882555233842344362162520787653763315998320477054896263434901090 \text{ and}$$

$$B = 292134647723778159914962540207823140335126344033210264225285566059959 \\ 956372273970360496308095920463669048323126517085888353814417281244783.$$

Since B is composite, we must go out to $m = 11$ to get the first known such prime:

$$p = 1193922124532829817155007545891716229062294355364062379437382238147944$$

$$1024110982077932762096079708192333239719602498611135957673315065156773.$$

The following corollaries are consequences of adding the Brier number condition.

Corollary 1. *The arithmetic progression, $Am + B$ for any nonnegative integer m , contains infinitely many primes which are Sierpiński numbers where*

$$36893488147419103230m + 10691053702625738573.$$

Corollary 2. *The arithmetic progression, $Am + B$ for any nonnegative integer m , contains infinitely many primes which are Riesel numbers where*

$$A = 3189762509404991434026267111935081578564848099075598418737381122 \text{ and} \\ B = 1464481329052452484012410951111069258728979724777952402192519099.$$

Corollary 3. *The arithmetic progression, $Am + B$ for any nonnegative integer m , contains infinitely many primes which are Brier numbers where*

$$A = 19613577555635811210751525911492599150619172808998516420233576073634 \\ 048313995204010 \text{ and} \\ B = 76099738321991043329005201186973924664733439211950059278085420869117 \\ 02037440072483.$$

These corollaries provide arithmetic progressions of integers with these properties different from those mentioned previously (see [22], [20], and [8]). All of the above results can be extended using a result of Shiu (see [21]) or a stronger result of Maynard (see [18]) just as was done in [11]. We will demonstrate these extensions using the Theorem 2 of this paper.

Corollary 4 (Shiu Corollary). *For every positive integer k , there exist k consecutive primes each of which is a $[0, \infty)$ -translated unit sensitive prime and a Brier number.*

Corollary 5 (Maynard Corollary). *For every positive integer k , there exist k consecutive primes $p_\ell, p_{\ell+1}, \dots, p_{\ell+k-1}$, each of which is both a $[0, \infty)$ -translated unit sensitive prime and a Brier number.*

2. Construction

To prove Theorem 1 and Theorem 2 of this paper, an arithmetic progression $Am + B$ will need to be constructed that satisfies two main criteria; $Am + B$ is a $[0, \infty)$ -translated unit sensitive number and $Am + B$ is a Brier number.

To start the construction, fix A and B as positive integers. In order for the arithmetic progression to contain infinitely many primes, fix A and B to be relatively prime (Dirichlet). In addition to these conditions, A and B will be constructed such that the following criteria are met.

- (1) For all integers $n \in [0, 9]$, each integer in the set

$$\mathcal{D}_n = \mathcal{D}_n(A, B) = \{(Am + B) \cdot 10^d + n : m, d \text{ are nonnegative integers, } d \neq 0\}$$

is divisible by at least one prime, p , such that p divides A .

(2) For all integers $x \in \{-3, -2, -1, 1, 2, 3, 4, 5, 6\}$, each integer in the set

$$\mathcal{D}_x = \mathcal{D}_x(A, B) = \{(Am + B) + x : m \text{ is nonnegative integer}\}$$

is divisible by at least one prime, p , such that p divides A .

(3) Each integer in the set

$$\mathcal{S} = \mathcal{S}(A, B) = \{(Am + B) \cdot 2^k + 1 : m, k \text{ are nonnegative integers}\}$$

is divisible by at least one prime, p , such that p divides A .

(4) Each integer in the set

$$\mathcal{R} = \mathcal{R}(A, B) = \{(Am + B) \cdot 2^k - 1 : m, k \text{ are nonnegative integers}\}$$

is divisible by at least one prime, p , such that p divides A .

One can deduce that criterion (3) would ensure $Am + B$ is a Sierpinski number and criterion (4) would ensure $Am + B$ is a Riesel number so long as $Am + B$ is odd. Thus, the combination of criteria (3) and (4) would ensure $Am + B$ is a Brier number as long as $Am + B$ is odd.

One can also deduce that criterion (1) would ensure $Am + B$ is a $[1, \infty]$ -translated unit number. While not as obvious, one can deduce that criterion (2) would ensure that $Am + B$ is unit sensitive (or 0-translated unit sensitive) so long as $Am + B \equiv 0 \pmod{10}$ (or $Am + B \equiv 1 \pmod{2}$ and $Am + B \equiv 3 \pmod{5}$) as Appendix Table 7 suggests). To help clarify the deduction from criterion (2), note that the initial conditions in this criterion A and B are designed under the assumption $Am + B \equiv 3 \pmod{10}$ for any nonnegative m .

Table 4: $Am + B + x$ under the conditions in criterion (2)

x	$Am + B + x \pmod{10}$
-3	0 (mod 10)
-2	1 (mod 10)
-1	2 (mod 10)
1	4 (mod 10)
2	5 (mod 10)
3	6 (mod 10)
4	7 (mod 10)
5	8 (mod 10)
6	9 (mod 10)

Using Table 4, we can see that all possible unit translations of $Am + B$ are accomplished by criterion (2) skipping $x = 0$ so that $Am + B + x = Am + B$ can be

prime. Thus, criterion (2) corresponds to $Am + B$ being unit sensitive. Therefore, criteria (1) and (2) would ensure that $Am + B$ is $[0, \infty)$ -translated unit sensitive.

The reason for separating the criteria for $Am + B$ to be $[0, \infty)$ -translated unit sensitive into two different criteria is worth noting as well. Consider the prime 83. To check if 83 is 1-translated unit sensitive, one checks that

$$830, 831, 832, 833, 834, 835, 836, 837, 838, \text{ and } 839$$

are all composite. In other words, since $83 \cdot 10^1 = 830 \equiv 0 \pmod{10}$, we can express the list of integers that need to be composite as $830 + n$ such that n is any integer in $[0, 9]$. The same can be done for any positive integer d . Just as criterion (1) indicates, this would mean for 83 to be d -translated unit sensitive, $83 \cdot 10^d + n$ must be composite for all integers $n \in [0, 9]$ since $83 \cdot 10^d \equiv 0 \pmod{10}$. However, to check if 83 is unit sensitive ($d = 0$) we want to check that

$$80, 81, 82, 84, 85, 86, 87, 88, \text{ and } 89$$

are all composite. Since $83 \not\equiv 0 \pmod{10}$, it is not sufficient to check if $83 \cdot 10^0 + n$ is composite for all integers $n \in [0, 9]$. This would be confirming that

$$83, 84, 85, 86, 87, 88, 89, 90, 91, \text{ and } 92$$

are all composite not the intended list of integers. Since $83 \equiv 3 \pmod{10}$, criterion (2) considers the intended list of integers.

3. Coverings

This section will discuss the construction of the coverings that satisfy criteria (1) - (4) as well as the initial conditions (A and B positive, relatively prime integers).

Before coverings are constructed, there are some cases within each criterion that need no coverings. First, consider criterion (1). Notice that when n is even, $(Am + B) \cdot 10^d + n$ is even for all positive d . Similarly, when $n = 5$, $(Am + B) \cdot 10^d + n$ is divisible by 5. Thus, coverings are only necessary for $n \in \{1, 3, 7, 9\}$ to satisfy criterion (1).

Now consider criterion (2) where $Am + B \equiv 3 \pmod{10}$. Similar to the above, so long as x is odd, $(Am + B) + x$ is even. Also, if $x = 2$, $(Am + B) + x$ is divisible by 5. Thus coverings are only necessary for $x \in \{-2, 4, 6\}$ to satisfy criterion (2).

The first covering will be for criterion (1) when $n = 1$ or $n = 7$. Consider $d \equiv 0 \pmod{1}$ with the prime 3. This congruence class contains all integers. In other words, since $(Am + B) \cdot 10^d + n \equiv (Am + B) + n \pmod{3}$ for any nonnegative integer d , the only congruence class in the covering can be $0 \pmod{1}$. Let

$$A \equiv 0 \pmod{3} \text{ and } B \equiv 2 \pmod{3}.$$

Then

$$Am + B + n \equiv 2 + n \pmod{3}.$$

Thus, $(Am + B) \cdot 10^d + n$ is divisible by 3 and composite when $A \equiv 0 \pmod{3}$ and $B \equiv 2 \pmod{3}$ so long as $n \equiv 1 \pmod{3}$ as well. the only n remaining that need a covering and are $n \equiv 1 \pmod{3}$ are $n = 1$ and $n = 7$ (as seen in Appendix 8). Therefore, the only remaining n from criterion (1) that need a covering are $n = 3$ and $n = 9$.

Now that $A \equiv 0 \pmod{3}$ and $B \equiv 2 \pmod{3}$, when $x = -2$ or $x = 4$ in criterion (2),

$$(Am + B) + x \equiv 2 + x \equiv 0 \pmod{3}$$

and, thus, composite. Therefore, only $x = 6$ remains to satisfy from criterion (2).

There will be other choices made for A and B according to covering systems that will be used in covering systems from other criteria as seen in Table 5. For example, from Table 5 one can see that choice of $Am + B \equiv 1 \pmod{7}$ ensures that when $x = 6$ in criterion (2), $(Am + B) + x \equiv 0 \pmod{7}$ and, thus completing criterion (2). It is noteworthy that when Table 5 indicates $Am + B \equiv a \pmod{p}$ (such as 3 (mod 11)) then $A \equiv 0 \pmod{p}$ and $B \equiv a \pmod{p}$ which in turn gives $Am + B \equiv a \pmod{p}$.

Table 5: All Reused Congruence Classes on $Am + B$ Part 1

2 (mod 3)	1 (mod 7)
(1) $n = 1$ and $n = 7$	(1) $n = 9, d \equiv 5 \pmod{6}$
(2) $x = -2$ and $x = 4$	(2), $x = 6$
(3) $k \equiv 0 \pmod{2}$	(4), $k \equiv 0 \pmod{3}$
(4) $k \equiv 1 \pmod{2}$	-

Table 5: All Reused Congruence Classes on $Am + B$ Part 2

3 (mod 11)	3 (mod 10)
(1) $n = 3, d \equiv 1 \pmod{2}$	(2)
(4), $k \equiv 2 \pmod{10}$	(3), $k \equiv 3 \pmod{4}$

With criterion (2) completed, as indicated by Table 5, coverings for criterion (1) with $n = 3$ and $n = 9$, criterion (3), and criterion (4) remain.

To complete criterion (1) for $n = 3$, consider the congruence class in the covering $d \equiv a \pmod{b}$. We choose b such that the order of 10 modulo the prime p divides b such that $10^b \equiv 1 \pmod{p}$. Then

$$(Am + B) \cdot 10^d + 3 \equiv (Am + B) \cdot 10^{a+be} + 3 \equiv (Am + B) \cdot 10^a + 3 \pmod{p}.$$

Letting p divide A , if we want $(Am + B) \cdot 10^d + 3$ to be composite by being divisible by p , then

$$(Am + B) \cdot 10^a + n \equiv 0 \pmod{p} \Rightarrow B \equiv \frac{-3}{10^a} \pmod{p}.$$

So long as $p \neq 2, \neq 5$ and $B \not\equiv 0 \pmod{p}$, B exists and A and B remain relatively prime such that $(Am + B) \cdot 10^d + 3$ is divisible by p . For criterion (1) with $n = 9$, criterion (3), and criterion (4), the process is similar as seen in Table 6.

Table 6: Congruence on B given $a \pmod{b}$ and prime p for each remaining criterion

Criterion	Congruence on B	Conditions
(1), $n = 3$	$B \equiv (-3/10^a) \pmod{p}$	$p \neq 2, p \neq 5, B \not\equiv 0 \pmod{p}$
(1), $n = 9$	$B \equiv (-9/10^a) \pmod{p}$	$p \neq 2, p \neq 5, B \not\equiv 0 \pmod{p}$
(3)	$B \equiv (-1/2^a) \pmod{p}$	$p \neq 2$ and $B \not\equiv 0 \pmod{p}$
(4)	$B \equiv (1/2^a) \pmod{p}$	$p \neq 2$ and $B \not\equiv 0 \pmod{p}$

Using the constructions highlighted in Table 6, so long as no primes are repeated (besides the few mentioned previously in Table 5), each congruence class will relate to a congruence class on $Am + B$. Now that we have defined the method for the coverings, please see the Appendix to see all coverings (Tables 7-12). Before we can explicit construct A and B from the coverings, we must explain Appendix Table 7. In order for criteria (2) to work, we required $Am + B \equiv 3 \pmod{10}$. However, a congruence class in the covering system for criteria (3) requires that $Am + B \equiv 3 \pmod{5}$ and, to be Sierpiński and Riesel, $Am + B$ must be odd. In order to satisfy these conditions, we will require $Am + B \equiv 1 \pmod{2}$ and $Am + B \equiv 3 \pmod{5}$ (just as Appendix Table 7 indicates) which is equivalent to $Am + B \equiv 3 \pmod{10}$.

Let

$$\mathcal{P}_{\mathcal{D}_n} = \{p : p \text{ is a prime associated to a covering for } n \text{ from criterion (1)}\}$$

and

$$\mathcal{B}_{\mathcal{D}_n} = \{B \pmod{p} : B \text{ is constructed according to Table 6 for prime } p\}.$$

Similarly, define $\mathcal{P}_{\mathcal{D}_x}$ and $\mathcal{B}_{\mathcal{D}_x}$, $\mathcal{P}_{\mathcal{S}}$ and $\mathcal{B}_{\mathcal{S}}$, and $\mathcal{P}_{\mathcal{R}}$ and $\mathcal{B}_{\mathcal{R}}$. Let

$$\mathcal{P} = \left(\bigcup_{\text{all } n \text{ in } \{1,3,7,9\}} \mathcal{P}_{\mathcal{D}_n} \right) \cup \left(\bigcup_{\text{all } x \text{ in } \{-2,4,6\}} \mathcal{P}_{\mathcal{D}_x} \right) \cup \mathcal{P}_{\mathcal{S}} \cup \mathcal{P}_{\mathcal{R}} \cup \{2\}$$

and

$$\mathcal{B} = \left(\bigcup_{\text{all } n \text{ in } \{1,3,7,9\}} \mathcal{B}_{\mathcal{D}_n} \right) \cup \left(\bigcup_{\text{all } x \text{ in } \{-2,4,6\}} \mathcal{B}_{\mathcal{D}_x} \right) \cup \mathcal{B}_{\mathcal{S}} \cup \mathcal{B}_{\mathcal{R}} \cup \{1 \pmod{2}\}.$$

We know that each prime must divide A so we get

$$A = \prod_{p \in \mathcal{P}} p.$$

Since all the prime moduli in $\mathcal{B}$ are unique with unique moduli, the Chinese Remainder Theorem ensures that there is some positive integer a such that

$$B \equiv a \pmod{A}$$

and Dirichlet's Theorem completes all that is need for Theorem 2. To complete Theorem 1, criteria (3) and (4) are not needed.

4. Note on Computational Methods

The authors would like to quickly note the methods used for finding the congruence classes for the coverings and the associated primes as well as how the coverings can be confirmed.

To find the associated primes from the congruence classes in the coverings, Magma V2.25-7 was used for factoring $\Phi_b(10)$. Given a congruence class in the covering, say $a \pmod{b}$, the prime p was found by fully factoring $\Phi_b(10)$ to its prime factorization and choosing p such that p had not already been chosen previously.

One can confirm a covering by finding the least common multiple of all the moduli and confirming every integer from 0 to one less than the least common multiple is satisfied by at least one of the congruences in the covering. For this paper, this was sufficient when using Maple 2019. For larger coverings, this can become an issue. For a different method of covering verification please see Filaseta, Juillerat, and the first author.

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Appendix

To aid in verifying the computations in this paper, all the data in this appendix can be found in [17] as lists suitable for computations. This appendix begins with Table 7 which illustrates which primes were needed in satisfying criteria (2). This table also includes the associated congruence classes B in the arithmetic progression $Am + B$ must satisfy with the primes each dividing A .

Tables 8-10 contain the congruence classes used to complete the coverings for $n = 1$ and $n = 7$, $n = 3$, and $n = 9$ respectively along with the associated primes and congruence classes for B to satisfy. The careful combination of these tables completes criteria (1).

Tables 11 and 12 contain the congruence classes used to complete the coverings for criteria (3) and (4) with the same structure as Tables 8-10. The careful combination of these tables completed the criteria for the arithmetic progression $Am + B$ to be made of Brier numbers.

There are several entries in the tables of this appendix that are notated with *. This is used to note a prime and associated congruence on B that was reused. A prime was reusable in another covering so long as the congruence class for B to satisfy remained the same. For example in Table 8 the prime 3 is used so that $B \equiv 2 \pmod{3}$. This prime is used again in both Table 11 and Table 12 with $B \equiv 2 \pmod{3}$.

Table 7: Covering Systems for Criteria (2)

x	p	$B \pmod{p}$
-	2^*	$1 \pmod{2}$
-	5^*	$3 \pmod{5}$
$x = -2$ and $x = 4$	3^*	$2 \pmod{3}$
$x = 6$	7^*	$1 \pmod{7}$

Table 8: Covering System for $n = 1$ and $n = 7$

Congruence Classes	p	$B \pmod{p}$
$d \equiv 0 \pmod{1}$	3^*	$2 \pmod{3}$

Table 9: Covering System for $n = 3$

Congruence Classes	p	$B \pmod{p}$
$d \equiv 1 \pmod{2}$	11^*	$3 \pmod{11}$
$d \equiv 2 \pmod{4}$	101	$3 \pmod{101}$
$d \equiv 4 \pmod{8}$	73	$3 \pmod{73}$
$d \equiv 0 \pmod{8}$	137	$134 \pmod{137}$

Table 10: Covering System for $n = 9$

Congruence Classes	p	$B \pmod{p}$
$d \equiv 1 \pmod{3}$	37	$25 \pmod{37}$
$d \equiv 2 \pmod{6}$	13	$12 \pmod{13}$
$d \equiv 5 \pmod{6}$	7^*	$1 \pmod{7}$
$d \equiv 6 \pmod{9}$	333667	$324667 \pmod{333667}$
$d \equiv 3 \pmod{18}$	19	$4 \pmod{19}$
$d \equiv 12 \pmod{18}$	52579	$43588 \pmod{52579}$
$d \equiv 9 \pmod{27}$	757	$514 \pmod{757}$
$d \equiv 18 \pmod{27}$	440334654777631	$440325654777631 \pmod{440334654777631}$
$d \equiv 0 \pmod{54}$	70541929	$70541920 \pmod{70541929}$
$d \equiv 27 \pmod{54}$	14175966169	$9 \pmod{14175966169}$

Table 11: Covering System for Sierpinski

Congruence Classes	p	$B \pmod{p}$
-	2^*	1 (mod 2)
$k \equiv 0 \pmod{2}$	3^*	2 (mod 3)
$k \equiv 3 \pmod{4}$	5^*	3 (mod 5)
$k \equiv 5 \pmod{8}$	17	9 (mod 17)
$k \equiv 9 \pmod{16}$	257	129 (mod 257)
$k \equiv 17 \pmod{32}$	65537	32769 (mod 65537)
$k \equiv 1 \pmod{64}$	641	320 (mod 641)
$k \equiv 33 \pmod{64}$	6700417	3350209 (mod 6700417)

Table 12: Covering System for Riesel

Congruence Classes	p	$B \pmod{p}$
-	2^*	1 (mod 2)
$k \equiv 1 \pmod{2}$	3^*	2 (mod 3)
$k \equiv 0 \pmod{3}$	7^*	1 (mod 7)
$k \equiv 0 \pmod{5}$	31	1 (mod 31)
$k \equiv 2 \pmod{10}$	11^*	3 (mod 11)
$k \equiv 1 \pmod{15}$	151	76 (mod 151)
$k \equiv 11 \pmod{45}$	631	57 (mod 631)
$k \equiv 26 \pmod{45}$	23311	11446 (mod 23311)
$k \equiv 86 \pmod{90}$	18837001	16 (mod 18837001)
$k \equiv 8 \pmod{20}$	41	37 (mod 41)
$k \equiv 18 \pmod{40}$	61681	61677 (mod 61681)
$k \equiv 38 \pmod{80}$	4278255361	4278255357 (mod 4278255361)
$k \equiv 78 \pmod{160}$	414721	414717 (mod 414721)
$k \equiv 158 \pmod{160}$	44479210368001	4 (mod 44479210368001)
$k \equiv 4 \pmod{30}$	331	269 (mod 331)
$k \equiv 14 \pmod{60}$	61	39 (mod 61)
$k \equiv 44 \pmod{60}$	1321	807 (mod 1321)